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Individual-Level Effects of a Digital Behavioural Treatment for Chronic Pain: Proof-of-Concept of a Single-Case Experimental Design Study

Haya Al Sharaa[1,2] | Sara Laureen Bartels[1,3] | Afra S. Taygar[1] | Linnéa Engman[1,2,4] | Suzanne Petersson[1,5] | Ida Flink[6] | Katja Boersma[4] | Lance M. McCracken[7] | Laura Simons[8] | Johan W. S. Vlaeyen[9,10] | Patrick Onghena[11] | Rikard K. Wicksell[1,2]

[1]Department of Clinical Neuroscience, Karolinska Institutet, Stockholm, Sweden | [2]Pain Clinic, Capio St. Göran's Hospital, Stockholm, Sweden | [3]Department of Psychiatry and Neuropsychology and Alzheimer Centrum Limburg, Mental Health and Neuroscience Research Institute, Maastricht University, Maastricht, the Netherlands | [4]School of Behavioural, Social and Legal Sciences, Örebro University, Örebro, Sweden | [5]Department of Medicine and Optometry, Linnaeus University, Kalmar, Sweden | [6]Department of Social and Psychological Studies, Karlstad University, Karlstad, Sweden | [7]Division of Clinical Psychology, Department of Psychology, Uppsala University, Uppsala, Sweden | [8]Department of Anesthesiology, Perioperative and Pain Medicine, Stanford University School of Medicine, Palo Alto, California, USA | [9]Experimental Health Psychology, Maastricht University, Maastricht, the Netherlands | [10]Health Psychology Research Group, KU Leuven, Leuven, Belgium | [11]Methodology of Educational Sciences Research Group, Faculty of Psychology and Educational Sciences, KU Leuven, Leuven, Belgium

Correspondence: Haya Al Sharaa (haya.al.sharaa@ki.se) | Sara Laureen Bartels (sara.bartels@ki.se)

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ABSTRACT

Background: Chronic pain affects 20%–30% of the population worldwide, leading to significant distress, disability and financial burden. Pain management strategies focusing on pain reduction have shown limited effects on functioning; however, behavioural treatments aimed at enhancing resilience have demonstrated strong empirical support. Digital solutions offer new opportunities for delivering evidence-based treatments, but evaluation at the individual level is needed. The aim of this study is to examine individual-level treatment effects of a digital behavioural treatment for chronic pain in a heterogeneous sample.

Methods: A study with a single-case experimental design (SCED) was conducted with participants ($N = 11$) experiencing chronic pain (> 3 months) recruited through healthcare. Participants were randomised at baseline (5–10-day A-phase) and completed a 6-module digital treatment based on learning theory and well-established theories applied to chronic pain (6–8-week B-phase), with weekly therapist contact. Digital diaries, prompted twice daily, tracked psychological flexibility and acceptance, pain-related functioning, pain intensity and well-being. Data were analysed using visual analysis and effect size calculations.

Results: $N = 11$ enrolled and data from $n = 10$ were analysed ($n = 1$ refused digital diary, $n = 2$ partial completers, $n = 8$ full completers). Pain profiles varied (e.g., chronic migraine, fibromyalgia, lower back pain, etc.). Several participants benefited from the treatment, though results varied across individuals and across outcomes.

Conclusion: The digital behavioural treatment showed promise in addressing diverse pain profiles and associated functioning. The variability in responses highlights the benefit of using SCED to explore individual-level effects, thus offering a methodological proof-of-concept. Findings support further development, including tailoring to match individual needs.

Haya Al Sharaa and Sara Laureen Bartels have contributed equally to this work.

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Significance Statement: This proof-of-concept study provides support for the utility of digital behavioural interventions and individual-level evaluation of treatment effects, highlighting the potential of personalised pain treatments. The findings contribute to the growing body of support for digital solutions as effective and accessible approaches to improve functioning and resilience for people with diverse pain experiences.

1 | Introduction

Chronic pain, typically defined as pain that persists or recurs for more than three months (Treede et al. 2019), affects 20%–30% of the population worldwide (Breivik et al. 2006; Zimmer et al. 2022), causing significant distress, disability and financial burden (Cohen et al. 2021). This condition is highly heterogeneous, impacting men and women of all ages (Breivik et al. 2006), with variations in pain intensity and reported disability (Adams et al. 2006). Generally, pain management focused on reducing pain and distress has shown limited effects on functioning and health-related quality of life (Wicksell and Vowles 2015). In contrast, behavioural approaches aimed at enhancing resilience to improve functioning have strong empirical support (Lai et al. 2023; Martinez-Calderon et al. 2024; Veehof et al. 2016). Resilience, defined as the ability to overcome adversity while maintaining effective functioning (Sturgeon and Zautra 2013), enables individuals to engage in meaningful activities that promote functioning despite pain and distress (Goubert and Trompetter 2017). Also, resilience may be a more important predictor of current and future health than the level of pain (Gentili et al. 2019; Zetterqvist et al. 2017). These findings highlight the importance of enhancing resilience through behavioural interventions rather than merely focusing on the reduction in pain and distress, to improve long-term health and functioning in people with chronic pain.

Furthermore, considering the complex interactions of genetic, demographic and psychosocial factors in pain experience (Fillingim 2017), there is a critical need for individual-level analysis when developing personalised treatments. However, currently dominating group-level analyses, such as randomised controlled trials (RCTs), lack the ability to detect heterogeneous treatment effects (Kent et al. 2010). RCTs typically yield group-level means while assuming population homogeneity and ergodicity—conditions rarely met in heterogeneous clinical populations (Zuidersma et al. 2020; Molenaar 2004), such as chronic pain (Adams et al. 2006). In contrast, single-case experimental design (SCED) allows for rigorous within-person analysis (Zuidersma et al. 2020), enabling individual-level evaluations of treatment effects. SCED has been used successfully to evaluate individual-level effects of behavioural treatments in chronic pain patients, such as for chronic back pain (Flink et al. 2009), irritable bowel syndrome (Boersma et al. 2016) and vulvodynia (Engman et al. 2022).

Additionally, digital solutions provide a paradigm shift in healthcare (Bartels et al. 2024), improving access to evidence-based treatments (Marcu et al. 2022). When developed, evaluated and implemented rigorously, digital solutions can enhance care quality, for instance, through reducing inequity (Hadjiat 2023). Accordingly, the DAHLIA (Digital behaviourAl HeaLth for chronIc pAin) project aims at improving healthcare standards for chronic pain management (Bartels et al. 2022). A digital behavioural treatment was developed in accordance with the Medical Research Council's framework for developing and evaluating complex treatments (Skivington et al. 2021). The digital behavioural treatment was then tested in the first of a series of optimisation studies, using small cohorts to evaluate and refine the treatment. The present study aimed to examine individual-level treatment effects in a chronic pain sample consisting of individuals with various pain conditions. We aim to offer a methodological proof-of-concept for the utility of this individualised approach in evaluating the effects of a novel treatment. The following research questions guide the design and data collection processes:

1. What individual-level impact does the treatment have on measures of psychological flexibility and acceptance, indicators of pain-related functioning, as well as pain intensity and well-being? And more specifically:
   a. Towhat extent do treatment effects vary across participants?
   b. To what extent do treatment effects vary across outcomes?
2. What are the pre-post changes per individual in measures of psychological flexibility and acceptance, indicators of pain-related functioning, as well as pain intensity and well-being?

2 | Methods

2.1 | Study Design

The present study follows the previously published study protocol of the multi-phase DAHLIA project (Bartels et al. 2022). This study is the first iteration in a series of optimisation studies. In the present study, a sequential replicated and randomised SCED was used (Tanious et al. 2024). In SCEDs, each patient's dependent variable(s) (see Section 2.5.1) is assessed repeatedly over time, through baseline (A-phase) and treatment (B-phase), with each sequential case considered a replication (Morley 2018). Also, the RoBiNT Scale, a comprehensive tool designed to evaluate the risk of bias in single-case experimental designs (Tate et al. 2013), was used to guide and assess the design.

2.2 | Participants and Recruitment

Participants' inclusion criteria were: (i) age 18–65 years (working ages), (ii) pain for at least 3 months, (iii) ability to communicate fluently in Swedish and (iv) access to a computer/tablet and smartphone with internet connection (to receive emails, notifications/SMS and receive the treatment). Exclusion criteria were: (i) serious psychiatric comorbidity (such as suicidality), (ii) serious injury/illness requiring immediate treatment or

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investigation (or expected to worsen in the next 6 months), (iii) changes in medications during the past or upcoming 3 months and (iv) ongoing or previous psychosocial treatment during the past 6 months. Eligibility was checked by the clinical coordinators (licensed psychologists/psychotherapists: SP, LE).

Recruitment took place through healthcare units in two regions (Kalmar and Stockholm). The clinical coordinators (SP, LE) provided information about the study to healthcare professionals and clinic managers. Patients were informed about the study by the clinical coordinator or other treating healthcare professionals, such as other psychologists/psychotherapists or medical doctors. Additionally, individuals who had expressed interest during the previous phases of the DAHLIA project, namely focus groups, were contacted via email. Interested participants were provided with a digital information sheet and had 1 week to consider participation before a link to the digital informed consent sheet was shared via email.

2.3 | Procedure

Eligibility screening was performed over the phone by the clinical coordinators following a semi-structured interview guide outlining inclusion and exclusion criteria. All participants provided informed consent through REDCap. The informed consent clarified that participation was voluntary and that participants could withdraw from the study without providing a reason. Next, participants completed online questionnaires (see Section 2.5.2), a baseline assessment including sociodemographic information, and started the daily digital diary (see Section 2.5.1). Later, therapists provided treatment access and facilitated initiation of the treatment (see Section 2.4).

Participants also completed weekly evaluations and exit interviews (see Bartels et al. 2022 for details; data will be presented elsewhere). Questionnaires were completed at pre- and post-treatment, as well as at 3- and 6-month follow-ups. Digital diary items were also administered at 3- and 6-month follow-ups. For the present study, digital diary SCED data was used to examine baseline to treatment individual-level treatment effects (A–B design), while baseline and post-treatment questionnaire data provide descriptive information. Follow-up data will be presented elsewhere.

2.4 | Treatment

The treatment (in Swedish 'Leva med Smärta') is a guided digital health treatment to increase resilience to pain and distress by enhancing behavioural self-management skills to improve functioning. The structure and content of the treatment are based on learning theory and an integration of well-established models in the chronic pain field, namely the fear-avoidance (Vlaeyen et al. 2016) and psychological flexibility models (McCracken and Morley 2014). Accordingly, the treatment aims to build resilience by promoting behavioural skills that support daily functioning and well-being, also in the presence of potentially interfering pain and distress. The treatment further enhances the individual's capacity to engage in values-oriented behaviours despite pain and distress, which can be empowering and contribute to improved functioning.

The treatment was developed in a user-centred process, that is, Phase 0 (preparation) and Phase 1 (design) of the DAHLIA project, including the use of Patient Personas, focus groups with end-users (patients and therapists), and iterative prototype testing to ensure the treatment matched the needs and preferences of the target group. For a detailed description of the development process, see Taygar et al. (2025).

2.4.1 | Content and Structure

During the treatment, participants are instructed to engage with six structured modules over the course of 6–8 weeks. An overview of the treatment modules and content of the micro-session is provided in Table 1 below. The treatment content builds on conceptual models emphasising values-based exposure and acceptance of pain and distress to promote resilience, functioning and well-being in successfully and sustainably managing chronic pain (McCracken and Vowles 2014; Boersma et al. 2016). Each module consists of four micro-sessions, which participants complete at their own pace. Each micro-session takes approximately 15–20 min. In each micro-session, reflections and exercises are provided and aim to promote self-awareness as well as behavioural change. At the end of each module, participants are encouraged to note reflections they wish to discuss with their therapist. After each module is completed (four micro-sessions; approximately 1 week), a brief phone or video call contact (30 min) with a therapist was provided. Participants had the same therapist throughout the treatment and progressed through the treatment sequentially, with each new module being unlocked by the therapist after the post-module contact.

Additionally, the platform allows participants to send messages to their therapist. Participants can expect replies from their therapist typically on a weekly basis. Also, therapists can monitor progress in real time, viewing responses to exercises and reflections. If participants do not complete sessions as scheduled, therapists can send reminders to encourage continued engagement.

In the present study, three licensed therapists provided the treatment (100% female; aged 36–58; 7–33 years of clinical experience; 2–30 years of experience with chronic pain patients). Therapists were briefed by the research team or clinical coordinator on how to deliver the treatment and followed a treatment manual.

2.5 | Data Collection

2.5.1 | Digital Diary

The diary items proposed in the protocol (Bartels et al. 2022) were piloted in a single case observation design (SCOD) study (Bartels et al. 2025, in manuscript), discussed with the research team, healthcare professionals, and experts in the Experience Sampling Method, resulting in a revised list of items (Table 2). Furthermore, outcome measures were carefully selected considering existing guidelines and empirical support, that is, the IMMPACT (Initiative on Methods, Measurement and Pain Assessment in Clinical Trials) recommendations for outcome domains in chronic pain trials

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TABLE 1 | Overview of treatment content.

Module	Session	Content of the micro-sessions
1—*You and your pain*	1	Psychoeducation on recurring pain and psychological treatment of pain
	2	Psychoeducation on the interplay between psychological factors and pain
	3	The role of behaviour in connection to pain and vicious circles
	4	Functional analysis of behaviour in relation to long- and short-term consequences
2—*Where do you want to go?*	1	Introduction to values in different areas of life
	2	Life values in relation to family and close relationships
	3	Values in relation to friends and health
	4	Values in relation to work/education and leisure activities
3—*What do you want to achieve?*	1	Values in relation to goals and goals in relation to family and close relationships
	2	Goals in relation to friends and health
	3	Goals in relation to work/education and leisure activities
	4	Reflections on a life led in the direction of one's values
4—*Moving forward*	1	Taking steps towards goals based on values
	2	Taking difficult steps towards goals even when it is difficult
	3	Negative thoughts and emotions as part of change and strategies to handle catastrophic thoughts
	4	Using the support from others
5—*Increasing strength and resilience*	1	Building skills to handle difficult change—that is, decreasing avoidance behaviour through exposure
	2	Dissecting goals and steps into manageable parts
	3	Emotion regulation strategies for strong affects
	4	Acceptance and validating strategies for self-criticism
6—*The road ahead*	1	Creating detailed plans of steps to take in relation to goals
	2	Repeating actions to form new habits
	3	Setting up plan for maintaining progress after the treatment
	4	Reflections on treatment progress and planning for the future

(Dworkin et al. 2008). Moreover, addressing specific and targeted psychological processes is critical to optimising the effects of the intervention (Kazdin 2007). In the present study, the selected outcome measures align with the process-based therapy framework (Hayes and Hofmann 2017) and optimisation frameworks such as MOST [The Multiphase Optimisation Strategy, which emphasises iterative testing of mechanisms alongside outcomes to refine interventions effectively (Collins et al. 2007)]. Accordingly, the measured constructs were selected to capture psychological flexibility/acceptance (process outcomes), pain-related functioning treatment effects (primary outcomes), as well as pain intensity and general well-being (secondary outcomes). Following the design of daily diary studies such as Experience Sampling trials (Verhagen et al. 2016), a 7-point Likert scale was used for most diary items. Pain intensity was measured on a 10-point scale, and pain interference was measured using multiple-choice.

The m-Path smartphone app (Mestdagh et al. 2023) was used for data collection. Diaries were programmed to send prompts twice daily, at 12.00 (lunch diary) and 18.00 (evening diary). The diaries were available for 2 h, and reminders were sent after 1 h if the participant did not provide an answer. Participants were given the option to change the time of the diaries if needed.

At baseline (A-Phase), participants were randomised to 5–10 days of digital diary sampling. For practical and logistical reasons, treatment start dates were concurrently scheduled by the therapist in advance and communicated to the researcher. The diary sampling (A-Phase) was initiated to reflect the randomised baseline length, ascertaining that the diary was completed for the correct number of days before treatment (B-Phase) started. During treatment (B-Phase), the diary was initially set for 6 weeks and extended if participants engaged with the treatment for a longer time period.

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TABLE 2 | Diary items used in the present study.

	Construct	Question	Answer scale
Process outcomes	Pain avoidance	*During the morning/afternoon* ... I stopped doing things because of my pain	7-point numeric scale (1: not at all—7: very much)
	Engagement in meaningful activity	... I did things that are important to me	
	General avoidance	... I avoided doing things because of how I felt	
Primary outcomes	Pain self-efficacy	... I felt confident that I could do activities despite my pain	7-point numeric scale (1: not at all—7: very much)
	Pain catastrophising	... I worried about my pain getting worse or not	
	Pain interference	... my pain interfered with ...	Multiple-choice: – General activities – Mood – Walking abilities – Normal work (including housework) – Relationship with others – Enjoyment of life – Did not disturb
Secondary outcomes	Pain intensity	... my pain level was: (1–10)	10-point numeric scale (1: no pain—7: worse possible pain)
	Positive affect	...I felt happy, energetic or enthusiastic	7-point numeric scale (1: not at all—7: very much)
	Negative affect	... I felt sad, irritated, depressed or hopeless	
	Sleep (general)[a]	Last night, I generally slept well	
	Sleep (falling asleep)[a]	Last night, I had problems falling asleep	
	Sleep (waking up)[a]	Last night, I woke up frequently or too early	
	Stress	*During the morning/afternoon* ... I felt stressed	
	Fatigue	... I felt tired	
	Social support	... I felt supported by others	

Note: In the study, the items were presented in Swedish and translated into English for transparency.
[a]Only included in the lunch questionnaire.

2.5.2 | Questionnaires

Participants completed a set of questionnaires pre- and post-treatment, presented in Table 3 below. All questionnaires are validated in Swedish. To reduce respondent burden, a selection of the instruments reported in the protocol (Bartels et al. 2022) was used. The questionnaires were selected to capture psychological flexibility and acceptance (process outcomes), pain-related functioning treatment effects (primary outcomes), as well as pain intensity and general well-being (secondary outcomes). Questionnaires were selected based on existing guidelines (IMMPACT) and empirical support (Dworkin et al. 2008; Kazdin 2007; Hayes and Hofmann 2017; Collins et al. 2007).

2.6 | Analytic Approach

Data were exported from REDCap and m-Path. A Python script was used to calculate individual questionnaire scores. Graphs for digital diary data were created in Excel. Missing questionnaire or diary data were not imputed; an approach used in most SCED studies (Aydin 2024).

2.6.1 | Visual Analysis

Graphs of repeated, numerical, item ratings from the diary measures were created including trendlines to enable

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TABLE 3 | Pre- and post-questionnaires used in this study.

Focus	Variable	Instrument	References
Process outcomes	Pain acceptance	Chronic Pain Acceptance Questionnaire (CPAQ-8)	Rovner et al. (2014)
	Psychological flexibility	Multidimensional Psychological Flexibility Inventory (MPFI-24)	Sundström et al. (2023)
	Avoidance/Fusion	Psychological Inflexibility in Pain Scale (PIPS)	Wicksell et al. (2008)
Primary outcomes	Catastrophising	Symptom Catastrophising Scale (SCS)	Moore et al. (2018)
	Pain self-efficacy	Pain Self-Efficacy Questionnaire (PSEQ)	Nicholas et al. (2015)
	Work ability	Work Ability Index (WAI)	Zetterberg et al. (2023)
	Functioning	Brief Pain Inventory—Short Form (BPI-SF)	Mendoza et al. (2006)
Secondary outcomes	Depression	Patient Health Questionnaire (PhQ-9)	Hansson et al. (2009)
	Anxiety	Generalised Anxiety Disorder Assessment (GAD-7). 5-Dimension Questionnaire.	Spitzer et al. (2006)
	Perceived stress	Perceived Stress Scale (PSS)	Eklund et al. (2014)
	Insomnia	Insomnia Severity Index (ISI)	Dragioti et al. (2015)
	Health-related quality of Life	EuroQol 5-Dimension Questionnaire (EQ-5D)	Burström et al. (2001))

visual inspection, a recommended procedure for SCED data (Morley 2018). Specifically, visual analyses followed a guideline (Kratochwill et al. 2010) focused on changes from A- to B- phases in means, trends, variability, overlap of data across phases, immediacy of change and consistency of pattern across all data points. Although trend lines were generated in the analyses, they were omitted from the visual figures due to concerns that they may confuse readers when data do not follow a clear linear pattern (Normand and Bailey 2006). Also, the structure and scaling of graphs may influence how trends are perceived, and the inclusion of trend lines can result in misinterpretation (Kubina et al. 2023). Thus, the slopes of the trends were calculated, as quantifying trends ensures higher reliability in assessing trends among researchers (Kubina et al. 2023). Moreover, the effects of the treatment were, in general, considered clearer when several of these examined indicators show change, following literature (Kazdin 2024). However, visual analysis was assessed holistically, as significant changes in one examined indicator of change can be more meaningful than smaller changes in another (Kennedy 2005).

To conduct a visual analysis for the multiple-choice pain interference item (see Table 2), data were recoded as a dichotomous variable: 0 = 'Did not interfere' (originally 'Did not disturb') and 1 = 'Interfered' for all other choices (general abilities, mood, walking abilities, normal work, relationship with others and enjoyment of life). For example, if a participant reported pain interference with mood and walking abilities, their response was coded as 1 ('Interfered'), whereas if they selected 'Did not disturb', their response was coded as 0. The recoded data were then used to depict graphs showing changes in pain interference over time.

Additionally, separate weekly graphs were created to show how often each interference area (general abilities, mood, walking abilities, normal work, relationship with others and enjoyment of life) was reported. For example, if a participant reported pain interference with general abilities every day of Week 1, the graph would depict 100% interference with general abilities for Week 1. However, no formal visual analysis was performed due to the nature of the item.

2.6.2 | Effect Size Calculation

Two robust non-parametric quantitative approaches to estimate effect sizes in SCEDs were applied, using the R-based web application Single-case effect size calculator (Pustejovsky et al. 2023).

Non-overlap of All Pairs (NAP) compares data points in phase A with data points in phase B to determine overlap or non-overlap and results in a percentage of non-overlap (Vannest and Ninci 2015). Moreover, Tau-u combines non-overlap with Theil-Sen regression method to correct for baseline trends (Vannest and Ninci 2015; Brossart et al. 2018). The Theil-Sen regression is a robust, nonparametric approach that calculates the median of slopes for all pairs of data points, providing a slope estimate (Tarlow 2017). Tau-u was calculated when trend correction was considered necessary, namely, when Tau was significant in phase A (Brossart et al. 2018). Tau in phase A was calculated using the online tool Baseline Corrected Tau Calculator (Tarlow 2016).

As individual needs and setting affect the interpretation of effect size calculations, benchmarks should be viewed as tentative rather than definitive (Vannest and Ninci 2015). However, to facilitate the interpretation of the results, the following tentative ranges were used:

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- NAP: 0–0.65 weak, 0.66–0.92 moderate and 0.93–1.0 large or strong. Values 0.50–1.00 generally indicate nondeteriorating change (Parker and Vannest 2009).
- Tau-u: 0–0.20 weak, 0.2–0.6 moderate, 0.6–0.8 large and >0.8 very large. A positive sign generally indicates nondeteriorating change (Vannest and Ninci 2015).

2.6.3 | Assessing Change in Pre- and Post-Questionnaires

No formal analysis was conducted across pre- and post-questionnaires. To identify meaningful changes in pre- and post-questionnaires, a threshold of 0.5 standard deviation—derived from the variability within the group—was used in accordance with existing recommendations (Dworkin et al. 2008).

3 | Results

3.1 | Participant Flow and Characteristics

A total of $n=11$ participants (P) enrolled in the study, of which $n=9$ completed the treatment (P-4 withdrew in module four reporting being too busy/not finding the treatment helpful and P-9 after module two due to not finding the treatment helpful). One participant (P-11) completed the treatment but not the digital diary and is, thus, only included in the descriptives and pre- and post-questionnaires.

Participants were between 36 and 58 years old, and mainly women ($n=9$). All participants were working or studying during the time of inclusion except for P-7 and P-10 who were on sick leave. Pain profiles varied and included chronic migraine, fibromyalgia, lower back pain, undiagnosed pain or osteoarthritis. Participant characteristics, treatment status, length of the baseline period (A-Phase), and diary compliance rates are presented in Table 4.

3.2 | Within-Person Effects

Visual analysis and effect size calculations are summarised in Table 5 below. Overall, positive treatment effects were observed, with improvements for most participants in at least four outcomes each, except P-9 and P-10, who only showed improvements in one outcome.

The most substantial positive effects were seen in participants P-4, P-6 and P-7, with improvements across 12, 8 and 11 outcomes, respectively. More than half of the participants showed improvements in pain catastrophising and sleep pattern, namely 'waking up frequently/too early'.

Some deterioration was also observed, mainly in P-10 who exhibited negative effects in four outcomes. Moreover, visual analysis and effect size calculations were generally consistent but also showed some instances of inconsistency.

Moreover, participants showed a variation in the level of pain interference across the different areas, namely general avoidance, mood, walking abilities, normal work, relations with others and enjoyment of life. As seen in Figure 1, for some participants, such as P-1 and P-7, areas of interference appear to fluctuate in a similar manner. In contrast, other participants, like P-3 and P-10, did not display a similar pattern; instead, changes in the respective areas of interference seem to fluctuate independently. Hence, some participants experienced similar increases or decreases in areas of interference, whereas others showed more varied responses.

3.2.1 | Exemplary Individual-Level Results

To provide a detailed individual illustration of the treatment effects, graphs over digital diary items are shown for three participants. These participants represent different levels of improvement, that is, moderate (P-2), substantial (P-4) and small to none (P-10).

Graphs illustrating the digital diary data for P-2 are shown in Figure 2. No changes were observed for process outcomes. However, improvements were observed in all three primary outcomes. Positive outcomes were also observed for the secondary outcomes 'pain intensity', 'negative affects' and 'fatigue'.

Graphs illustrating the digital diary data for P-4 are shown in Figure 3. Positive changes were observed for all process outcomes and both 'pain catastrophising' and 'pain interference' in the primary outcomes. No changes were observed for the primary outcome 'pain self-efficacy'. Also, positive changes were observed for all secondary outcomes except for 'pain intensity' and 'social support'.

Graphs illustrating the digital diary data for P-10 are shown in Figure 4. No positive changes were observed for any of the process or primary outcomes. The only positive improvement for P-10 was observed for the secondary outcome 'sleep: waking up frequently/too early'. Also, a deterioration was observed for the primary outcome 'pain interference', as well as the secondary outcomes 'pain intensity', 'stress' and 'fatigue'.

3.3 | Pre-Post-Treatment Questionnaires

Pre- and post-questionnaire data are presented in Table 6 below. Several individuals showed meaningful changes (changes of 0.5 standard deviations) in process, primary and secondary outcomes. Notably, all treatment completers showed meaningful positive changes for the subscale pain intensity in the Brief Symptom Inventory—Short Form (BSI-SF). However, for the subscale fusion in the Psychological Inflexibility in Pain Scale (PIPS), no meaningful positive changes were observed, yet negative changes were observed for two participants.

4 | Discussion

This study evaluated a digital behavioural treatment based on well-established theories in the chronic pain field. This study is the first iteration in a series of optimisation studies to examine

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TABLE 4 | Participant characteristics, treatment status, length of A-Phase and diary compliance rate.

	Age, sex	Education, occupation	Primary pain symptomatic (years since pain first appeared)	Status	Length of A-phase	Diary compliance rate (A/B-phase)
P-1	48, female	College/university; working, part-time	Chronic migraine; Myalgia encephalomyelitis (40 years)	Treatment completer	6	53.73%
P-2	36, female	College/university; studying, part-time	Chronic migraine (10 years)	Treatment completer	9	70.67%
P-3	37, male	Highschool; working, full-time	Undiagnosed widespread pain (7 years)	Treatment completer	7	71.23%
P-4	41, male	College/university; working, full-time	Herniated disc; Osteoarthritis; Undiagnosed lower back pain (9 years)	Partial completer. Dropped out due to being too busy/not finding the treatment helpful	10	30.93%
P-5	58, female	Highschool; working, part-time	Fibromyalgia and osteoarthritis; several years (not further specified)	Treatment completer	9	69.23%
P-6	48, female	Highschool, applied degree; working, part-time	Complex regional pain syndrome; hEDS (35 years)	Treatment completer	5	76.47%
P-7	52, female	College/university; sick leave, permanent	Chronic migraine (10 years)	Treatment completer	5	62.30%
P-8	43, female	College/university; working part-time	Complex regional pain syndrome (6 years)	Treatment completer	9	52.66%
P-9	45, female	Highschool; working full-time	Fibromyalgia; osteoarthritis; migraine (12 years)	Partial completer. Dropped out due to not finding the treatment helpful	7	88.09%
P-10	45, female	College/university; sick leave	Nociceptive pain, back pain (7 years)	Treatment completer	10	95.14%
P-11	53, female	Highschool, applied degree; studying, part-time	Herniated disc with widespread pain (5–10 years)	Treatment completer[a]	NA[a]	NA[a]

[a]P-11 completed the treatment, however, did not want to do the digital diary data collection.

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TABLE 5 | Summary of SCED visual and effect size analysis results.

Outcome level	Construct	SCED analysis	P-1	P-2	P-3	P-4[a]	P-5	P-6	P-7	P-8	P-9[a]	P-10	# of positive effects across participants	# of negative effects across participants
Process outcomes	Pain avoidance	Visual	n	n	n	+	++	n	+	+	n	n	4	0
		NAP (Tau-U)	0.56 [f]	0.51	0.41	0.76 [f]	0.70	0.55	0.64 [f]	0.70	0.52	0.53 [f]	4	0
	Engagement in meaningful activities	Visual	n	n	−	+	n	+	n	n	n	n	2	1
		NAP (Tau-U)	0.58 [c]	0.50	0.34	0.69	0.49	0.68	0.40	0.45	0.53	0.51	2	1
	General avoidance	Visual	n	n	+	+	+	n	n	n	n	n	3	0
		NAP (Tau-U)	0.49 [f]	0.56	0.34	0.76 [c]	0.62	0.54	0.49 [f]	0.62	0.57	0.59	2	1
Primary outcomes	Pain self-efficacy	Visual	++	+	++	n	n	n	++	n	+	n	5	0
		NAP (Tau-U)	0.56 (0.16)[c]	0.60	0.64	0.53	0.47	0.57	0.61	0.57	0.67[c]	0.53[c]	5	0
	Pain catastrophising	Visual	++	+	n	++	n	+	++	+	n	n	6	0
		NAP (Tau-U)	0.52	0.57[f]	0.49	0.71[f]	0.45	0.81[f]	0.73 [f]	0.65	0.50 [f]	0.50 [f]	6	0
	Pain interference	Visual	n	+	n	++	n	++	+	+	NA	−	5	0
		NAP (Tau-u)	0.54	0.65	0.44	0.71	0.52	0.68	0.57	0.63	NA	0.29 (−0.44)	5	0
Secondary outcomes	Pain intensity	Visual	n	+	n	n	n	+	+	++	n	−	4	0
		NAP (Tau-U)	0.51	0.65	0.56	0.59	0.57	0.61	0.70	0.65 (0.25)	0.44	0.35 (−0.37)	4	0
	Positive affect	Visual	n	n	n	+	+	+	n	n	n	n	3	0
		NAP (Tau-U)	0.59	0.46	0.40	0.59	0.58	0.70	0.41	0.42	0.44	0.53 (0.02)	3	0
	Negative affect	Visual	++	+	n	+	n	+	n	n	n	n	4	0
		NAP (Tau-U)	0.58	0.67	0.41	0.64	0.47	0.59[f]	0.53	0.50	0.62[f]	0.48[f]	4	0
	Sleep: General	Visual	n	n	n	++	++	+	+	−	n	n	4	1
		NAP (Tau-u)	0.48	0.41	0.58	0.73	0.79	0.61	0.60	0.34	0.31	0.40 (−0.23)	4	2

(Continues)

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TABLE 5 | (Continued)

Outcome level	Construct	SCED analysis	P-1	P-2	P-3	P-4[a]	P-5	P-6	P-7	P-8	P-9[a]	P-10	# of positive effects across participants	# of negative effects across participants
	Sleep: Problems falling asleep	Visual	n	n	+	+	n	-	++	+	n	n	4	1
		NAP (Tau-u)	0.48	0.41	0.55	0.71[f]	0.44	0.31	0.61	0.62	0.41	0.36	4	1
	Sleep: Waking up frequently/ too early	Visual	+	n	++	++	+	n	+	n	n	++	6	0
		NAP (Tau-u)	0.74	0.44	0.82	0.82	0.76	0.57	0.87	0.61	0.45	0.65 (0.26)	6	0
	Stress	Visual	n	n	+	+	n	++	+	n	n	−	4	1
		NAP (Tau-u)	0.53	0.43 (−0.11)	0.52	0.72 (0.39)	0.50	0.62	0.59[f]	0.45 (−0.14)	0.50 (−0.13)	0.39	4	1
	Fatigue	Visual	n	+	+	+	n	n	+	n	−	−	4	2
		NAP (Tau-u)	0.50 (0.04)	0.62	0.46	0.61	0.51	0.70	0.78	0.62 (0.23)	0.42 (−0.21)	0.41	4	2
	Social support	Visual	n	n	−	n	n	n	+	+	n	n	2	1
		NAP (Tau-u)	0.52	0.65[c]	0.27	0.57	0.56	0.66[c]	0.63	0.66	0.63 [c]	0.54[c]	3	1
#of positive visual treatment effects within participant			4	6	6	12	5	8	11	6	1	1		
# of negative visual treatment effects within participant			0	0	2	0	0	1	0	1	1	4		

Note: Light green indicating small improvements. Green indicating medium to large improvements. Pink indicating small deterioration. Red indicating medium to large deterioration. NAP is interpreted according to tentative ranges by Parker and Vannest (2009), where scores around 0.50 typically suggest no change, with lower scores typically indicating potential deterioration and higher scores suggesting a potential treatment effect; however, other indicators of change play a role in interpreting scores. Tau-U is only presented if Tau in phase A is significant. Pain interference is not analysed for P9 due to very low item compliance. [a]Partial completer, [f]flooring effect, [c]ceiling effect.

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FIGURE 1 | Reported pain interference in specific areas per participant. No visual analysis was conducted due to the nature of the item. Percentages showing how often each participant reported each interference area (general abilities, mood, walking abilities normal work, relationship with others and enjoyment of life) were calculated first for baseline followed by week-by-week data. P9 was not included due to very low item compliance.

the treatment's effects on chronic pain management at the individual level. The primary objective was to explore individual-level effects of the treatment, with an emphasis on the variability in measures of psychological flexibility and acceptance, indicators of pain-related functioning, as well as pain intensity and general well-being.

4.1 | Overview of Findings

The pattern of results suggests that several participants showed small to moderate improvements in several outcomes following the digital behavioural treatment. As expected, results varied across individuals as well as across variables, both in questionnaire data and digital diary data. Hence, some participants showed positive changes in certain domains while others improved in other domains. In a few participants, certain instances of deterioration were observed.

4.2 | Correspondence With Previous Research in the Field

Existing research shows strong empirical support for treatments aimed at enhancing resilience (Martinez-Calderon et al. 2024; Lai et al. 2023; Veehof et al. 2016). Yet, psychological treatments have also shown adverse effects in some patients (Honkalampi et al. 2024), which may be linked to cognitive-emotional factors such as levels of depression, anxiety and cognitions about pain (Forden et al. 2024). Also, the variations in effects seen in the present study are consistent with previous individual-level research on chronic pain (Flink et al. 2009; Boersma et al. 2016; Norman-Nott et al. 2021; Engman et al. 2022). Our results, as well as existing research, thus indicate that treatment effects and change mechanisms may differ due to several individual factors such as pain duration and insomnia (Gentili et al. 2023). These factors may help explain why participants improved in different ways and showed effects in different domains. However, more knowledge about how such factors vary between participants and influence treatment outcomes is needed. Prospectively, qualitative data through conversations with patients about their own treatment effects could shed light on this question. Such individual variability emphasises the importance of individual-level analyses, as patients in clinics typically may not fit the group-based average description (Zuidersma et al. 2020). Individual-level analysis enables a more detailed and nuanced understanding of treatment effects (Epstein and Dallery 2022). Moreover, a personalised, process-based approach to chronic pain treatment, where treatments are matched to the individual's unique context and mechanisms of change, has been suggested as a promising direction in the field of chronic pain treatment (McCracken 2023). Thus, our findings support a growing shift in pain research towards more personalised process-based models of treatment (Hayes and Hofmann 2017). The use of SCED and digital diary data illustrates how moment-to-moment change can inform individualised care, aligning with current efforts to develop adaptive and process-based interventions (McCracken 2023).

4.3 | Methodological Considerations and Suggestions for Future Research

When interpreting the results for individual participants, the statistical and visual analyses were at times inconsistent. More

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FIGURE 2 | Digital diary data for P-2 (moderate effects).

specifically, low effect sizes, which suggest possible deterioration, contrasted with visual improvements that became apparent towards the end of the treatment, as seen in P-3 with the item 'general avoidance'. Additionally, for P-9, the limited number of data points for the item 'sleep: general' resulted in inconsistencies between the effect size calculations and visual analysis, as the sparse data made it difficult to draw reliable conclusions visually.

Such inconsistencies highlight that NAP scores alone are insufficient to draw conclusions about the effects of a treatment, as individual participants' needs, setting and goals affect how effect sizes are interpreted (Vannest and Ninci 2015). Also, high NAP scores are sometimes affected by ceiling or floor effects (Manolov and Tanious 2024). For P-6, a change occurred during baseline in the item 'social support', with scores reaching a ceiling effect that persisted throughout the treatment. Consequently, NAP does not accurately reflect the effects of the treatment itself, as changes occurred already during baseline. This points to a methodological challenge; more specifically, NAP's inability to account for the magnitude of the change seen in the treatment phase (Carlin and Costello 2022) and to consider the trend in baseline data (A-phase) (Dowdy et al. 2021).

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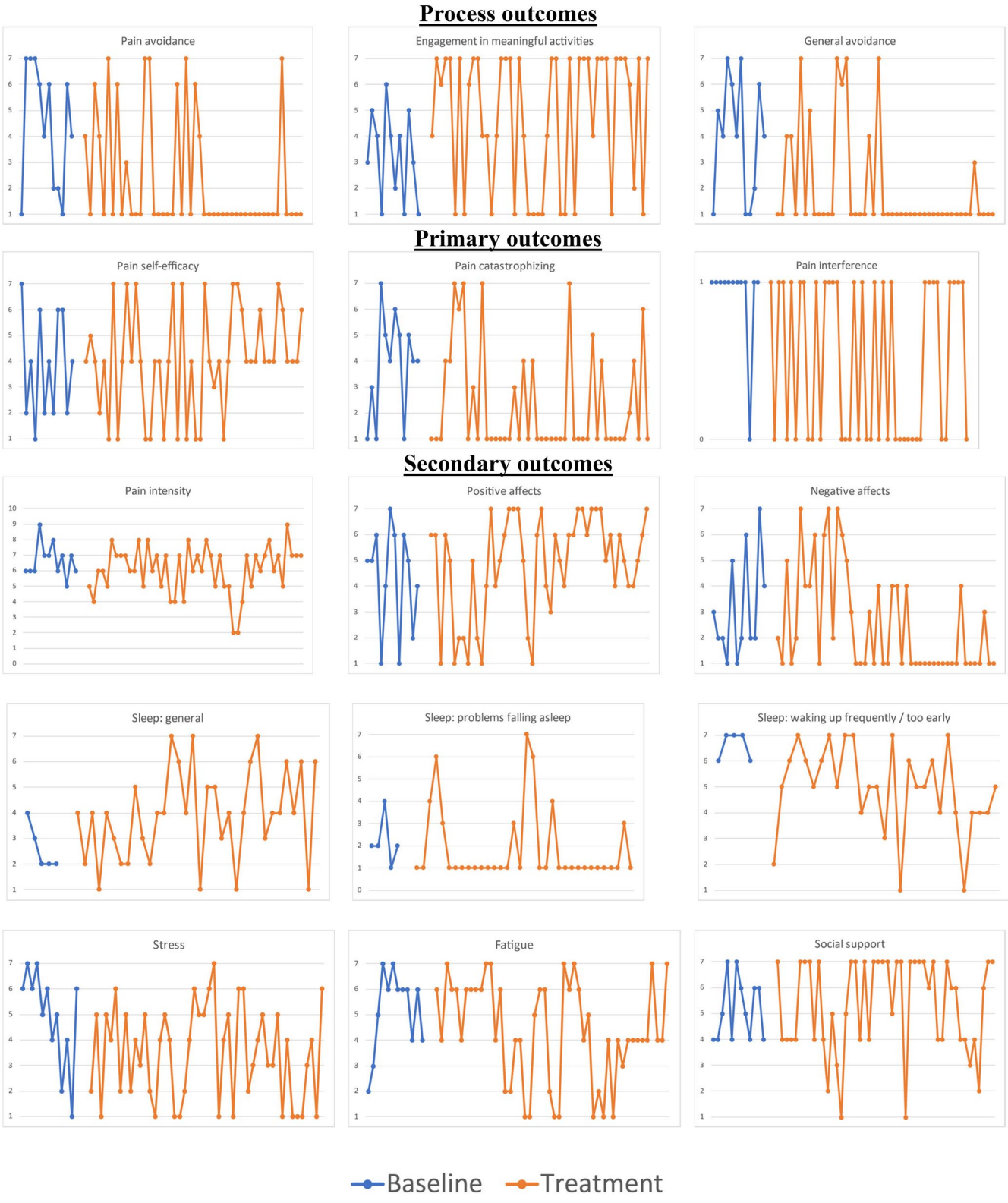


FIGURE 3 | Digital diary data for P-4 (substantial effects).

Hence, if used alone, effect size calculations such as NAP can potentially lead to incorrect conclusions about effects.

Notably, the changes in questionnaires from pre- to post-treatment do not fully align with changes observed in digital diary data. For instance, P-10 showed almost no improvements and some deterioration in the diary data, yet pre- and post-questionnaires show meaningful improvements in several outcomes. Also, whereas only four participants showed improvements in the digital diary item pain intensity, all treatment completers showed meaningful improvements in pain severity based on pre- and post-questionnaires. Hence, although both

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FIGURE 4 | Digital diary data for P-10 (small to no effects).

the questionnaires and digital diary constructs were selected to capture psychological flexibility and acceptance, pain-related functioning, pain intensity and general well-being, some discrepancies were still observed between momentary and psychometrically evaluated questionnaires. These discrepancies may be due to recall bias in the retrospective questionnaires (Stone et al. 2004). Additionally, natural fluctuations in symptoms may contribute to the explanation of discrepancies (Kratz et al. 2017; Schneider et al. 2012). Nonetheless, such discrepancies highlight the importance of combining momentary and psychometrically evaluated questionnaires for a comprehensive perspective on treatment effects.

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TABLE 6 | Scores for process, primary and secondary outcome measures assessed pre- and post-treatment ($n = 9$).

	Process outcomes (baseline|post-treatment)								Primary outcomes (baseline|post-treatment)					Secondary outcomes (baseline|post-treatment)				
	PIPS			CPAQ-8			MPFI-24				BPI-SF							
	Avoidance	Fusion	Total	Engagement in activities	Willingness to have pain	Total	Flexibility	Inflexibility	PSEQ	SCS	Pain severity	Pain interference	WAI	PHQ-9	GAD-7	PSS	ISI	EQ5D
P1	19|14	8|7	27|21	18|17	21|20	39|37	4.42|4.17	1.5|1.67	8|12	2|2	2.14|0	2.5|2.0	26.5|24.5	2|0	1|1	8|3	11|0	0.91|0.96
P2	37|21	15|14	52|35	9|10	17|16	26|26	3.83|4.67	2.42|2.33	5|10	7|4	8.14|0	4|1.5	21.5|26	12|9	5|3	18|13	4|4	0.67|1
P3	33|31	14|13	47|44	13|15	12|16	25|31	3.33|4.08	3.82|3.33	8|11	10|9	7.71|6	6.75|6.25	30|30	5|6	11|9	24|20	15|14	0.82|0.91
P4[a]	45|NA	20|NA	66|NA	18|NA	16|NA	34|NA	3.08|NA	3.83|NA	12|NA	12|NA	8.29|NA	6.25|NA	34|NA	14|NA	17|NA	22|NA	27|NA	0.77|NA
P5	35|33	19|19	54|52	7|10	11|12	18|22	3.52|3.58	2.92|2.83	4|9	9|9	7.57|5.86	5.75|4.75	23|22	12|10	13|8	24|26	15|15	0.73|0.82
P6	30|30	19|20	49|50	17|14	22|11	28|25	4.58|4.42	3.67|2.83	9|8	6|7	6|3.86	5.25|4.75	27|22	7|5	2|0	8|9	13|14	0.82|0.82
P7	36|31	22|26	58|57	6|14	8|9	14|23	3.92|4.83	2.5|2.83	5|7	8|5	4.29|0.29	3.75|0.75	30|27	4|1	2|3	6|8	1|1	0.96|0.91
P8	34|27	21|20	55|47	13|20	11|16	24|36	3.5|3.33	2.58|2.92	9|7	7|4	6.71|3.86	5.75|5.75	25.5|26	15|7	13|3	21|26	15|8	0.78|0.91
P9[a]	18|NA	10|NA	28|NA	19|NA	17|NA	36|NA	NA	NA	10|NA	2|NA	1.86|NA	4.75|NA	31|NA	6|NA	1|NA	2|NA	14|NA	0.91|NA
P10	36|12	15|11	51|23	4|20	15|23	19|43	5.42|5.5	2.42|1.67	2|10	8|3	6.57|4.71	4|5	20.5|19	11|10	5|2	28|16	23|23	0.87|0.77
P11	36|31	22|26	58|57	5|8	10|11	15|19	4.83|4.83	2.42|2.17	0|1	8|9	9|6	5.25|4.75	19|21	12|13	1|3	11|14	19|24	0.63|0.63

Note: Green indicating meaningful improvement. Red indicating meaningful deterioration. [a]P-4 and P-9 did not complete the post-treatment assessment due to drop-out.
Abbreviations: BSI-SF, Brief Pain Inventory—Short Form; CPAQ-8, Chronic Pain Acceptance Questionnaire; EQ5D, EuroQol 5-Dimension Questionnaire; GAD-7, Generalised Anxiety Disorder Assessment; ISI, Insomnia Severity Index; MPFI-24, Multidimensional Psychological Flexibility Inventory-24; PHQ-9, Patient Health Questionnaire-9; PIPS, Psychological Inflexibility in Pain Scale; PSEQ, Pain Self-Efficacy Questionnaire; PSS, Perceived Stress Scale; SCS, Symptom Catastrophising Scale; WAI, Work Ability Index.

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Furthermore, changes in process variables did not consistently coincide with changes in primary treatment outcomes. This temporal discrepancy is consistent with similar observations from previous ACT research. For instance, Scott et al. (2016) observed that cognitive fusion demonstrated greater change from pre-treatment to follow-up than from pre- to post-treatment, suggesting that psychological processes targeted in ACT may continue to develop after the treatment ends. Future research should explore post-treatment trajectories and assess mediators of change to better understand the mechanisms underlying therapeutic improvement.

Typically, baseline stability is recommended before introducing the treatment (Kratochwill and Levin 2014). In the present study, some participants exhibited high levels of fluctuations during the baseline phase. Longer baseline periods could help stabilise these measurements, ensuring a more reliable comparison between baseline and treatment phases (Kratochwill and Levin 2014), which should be considered in future iterations. However, pain-related functioning, as well as pain intensity and general well-being in the context of chronic pain are thought to fluctuate over time (Kratz et al. 2017; Schneider et al. 2012); thus, stability at baseline might not be achievable. To address this challenge, future studies could explore alternative approaches such as multilevel modelling. By modelling repeated measurements hierarchically, multilevel models can help distinguish between treatment-related changes and natural fluctuations (Baek et al. 2023). Furthermore, future research should examine whether the observed treatment effects are sustained over time, as long-term outcomes of the provided treatment remain uncertain.

Moreover, analysing the daily diary item 'Pain interference' posed challenges. Since a multiple-choice format was used, opposed to a numeric scale, visual analysis was limited to a dichotomous interpretation. Hence, fluctuations were less visible compared to other digital diary items. Additionally, no visual analysis was conducted for variations in the specific areas where pain interfered. As pain interference mediates effects on health-related quality of life (Varni et al. 2020), future iterations should consider using a visual analogue or a numeric scale to allow for more detailed trend analysis. Future studies may also compare different response scales for diary items (e.g., 5-, 7-, 10- or 100-point scales) to assess their sensitivity, usability and appropriateness in capturing momentary changes in the context of chronic pain.

Additionally, the sample characteristics, primarily well-educated, employed women, correspond with reported trends in digital health intervention research. Previous studies have shown that higher education and employment status are associated with greater engagement in digital treatments (Ruotsalainen et al. 2024). Moreover, women appear more likely to seek and adhere to such interventions, potentially due to higher health awareness and help-seeking behaviours (Boucher and Raiker 2024). These factors may influence participation and outcomes also in the present study, which suggests the importance of carefully considering recruitment and engagement strategies in future trials to broaden inclusion to generate a more diverse and representative sample.

Furthermore, this study achieved an internal validity score of 6/14 and an external validity score of 13/16 (see Table S1) as per the RoBiNT scale (Tate et al. 2013). Especially, blinding and objective monitoring of treatment adherence were not possible in this study, which implies a potential risk of bias that should be considered when interpreting the results. In contrast, the study's robust external validity suggests that the study provides valuable insight and meaningfully contributes to the development and individual-level assessment of digital behavioural treatments.

Finally, this study contributes to the field by demonstrating how SCEDs can be used to evaluate treatment effects, in line with personalised pain management and precision health. While group-level analyses such as RCTs are essential for validating the utility of the treatment on a population level, they often fail to capture individual-level changes (Kent et al. 2010). SCEDs enable a detailed understanding of how and when change occurs within individuals (Zuidersma et al. 2020). This study demonstrates that a SCED approach can detect heterogeneity in treatment responses and can be seen as a methodological proof-of-concept. As the chronic pain population is heterogenous, optimal effects may require individualised treatment. This implies the importance of SCED or similar approaches when conducting clinical trials, to understand individual variation in effects. Importantly, aggregating SCED data across individuals can also provide a broader understanding of what interventions work for whom (Kinney et al. 2025). Future research should thus continue building cumulative evidence using SCED methodologies, potentially through meta-analytic aggregation or advanced statistical modelling across replicated individual cases (Baek et al. 2023; Kinney et al. 2025).

5 | Conclusions

By examining patterns of improvement for each individual, this study provides insights into the potential of a digital behavioural treatment to address chronic pain and indicate the utility of individual-level analyses and personalised treatments. Results offer preliminary proof-of-concept for evaluating digital behavioural treatments, suggesting the relevance of a SCED approach in evaluations of individual-level effects. The treatment effects were heterogeneous and ranged from small to moderate. Although the results in general appear promising, the variability in effects warrants further development of the treatment, including tailoring the treatment to match individual needs. Such personalised pain management may improve effectiveness across a wider range of people with chronic pain. Further optimisation will be done using input from end-users as provided in weekly evaluations and exit interviews.

Author Contributions

H.A.S. conducted data analysis and was responsible for writing and editing the manuscript. S.L.B., as project manager, contributed to study planning, provided supervision, and was involved in manuscript writing. A.S.T. assisted with study planning and handled data collection. L.E. and S.P. were responsible for planning, coordinating with therapists, participant recruitment and providing treatment. I.F. supervised the study and contributed to planning. K.B., L.M.M., L.S. and J.W.S.V.

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served as advisors and reviewed the manuscript. P.O. provided methodological guidance and advised on data analysis. R.K.W., as principal investigator, led the research group, supervised the study, secured funding and contributed to planning. All authors discussed the results and commented on the manuscript.

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Conflicts of Interest

Co-authors S.P. and L.E. provided treatment but were not involved in data analysis. The remaining authors declare no conflicts of interest.

Data Availability Statement

The data supporting the findings of this study is available upon reasonable request from the corresponding authors.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Risk of Bias in N-of-1 Trials (2013). Green indicating full score (2 points), yellow indicating partial score (1 point), red indicating 0 points.

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